

Tesla Stock Price Prediction Using Deep Learning

1. Introduction:

Stock price prediction is one of the most challenging problems in financial forecasting due to market volatility and the influence of multiple external factors. Tesla Inc. is one of the most actively traded companies in the stock market, making it an ideal candidate for time-series forecasting analysis.

The objective of this project is to analyze Tesla stock price data and develop deep learning models capable of predicting future stock prices using historical observations.

2. Problem Statement:

The objective is to build predictive deep learning models that can forecast Tesla stock prices using historical stock market data.

The project focuses on:

- Data Cleaning
 - Exploratory Data Analysis
 - Feature Engineering
 - Deep Learning Model Development
 - Hyperparameter Tuning
 - Multi-Step Forecasting
 - Streamlit Deployment
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3. Dataset Description:

The Tesla stock dataset contains the following attributes:

- Date
- Open
- High
- Low
- Close
- Adj Close
- Volume

The analysis focuses primarily on Tesla's closing stock price.

4. Data Preprocessing:

The following preprocessing steps were performed:

4.1 Data Cleaning:

- Checked for missing values.
- No missing values were found.

4.2 Time-Series Missing Value Handling:

For time-series datasets, missing values should be handled carefully to preserve temporal relationships. Common techniques include forward filling, backward filling, and interpolation.

4.3 Date Conversion:

- Converted Date column to datetime format.
- Set Date column as index.

4.4 Data Scaling:

MinMaxScaler was applied to normalize stock prices between 0 and 1.

4.5 Sequence Creation:

A rolling window of 60 days was used to create input sequences for deep learning models.

5. Exploratory Data Analysis:

Several visualizations were created:

- Tesla Closing Price Trend
- Trading Volume Analysis
- 50-Day Moving Average
- 200-Day Moving Average
- Daily Returns Distribution
- Correlation Heatmap

These analyses helped identify stock trends, volatility, and feature relationships.

6. Feature Engineering:

The following features were generated:

- 50-Day Moving Average (MA50)
- 200-Day Moving Average (MA200)
- Daily Returns

These features helped improve understanding of Tesla's market behavior.

7. Deep Learning Models:

7.1 SimpleRNN:

Architecture:

- SimpleRNN Layer
- Dropout Layer
- SimpleRNN Layer
- Dropout Layer
- Dense Output Layer

7.2 LSTM:

Architecture:

- LSTM Layer
 - Dropout Layer
 - LSTM Layer
 - Dropout Layer
 - Dense Output Layer
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8. Model Optimization:

To improve training performance:

- EarlyStopping was implemented.
- ModelCheckpoint was used to save the best-performing model.

GridSearchCV was applied to evaluate multiple hyperparameter combinations.

Best Parameters:

- Units: 25
- Dropout Rate: 0.1

9. Model Evaluation:

Evaluation metrics:

- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- Mean Absolute Error (MAE)
- R^2 Score

9.1 SimpleRNN Performance:

- MSE: 310.29
- RMSE: 17.62
- MAE: 11.34
- R^2 Score: 0.9458

9.2 LSTM Performance:

- MSE: 554.95
- RMSE: 23.56
- MAE: 16.03
- R^2 Score: 0.9032

9.3 Best Model:

SimpleRNN achieved superior performance compared to LSTM.

10. Multi-Step Forecasting:

The trained model was used to forecast Tesla stock prices for:

- 1-Day Forecast
- 5-Day Forecast
- 10-Day Forecast

This demonstrates the model's ability to generate future stock price predictions over multiple forecasting horizons.

11. Streamlit Deployment:

A Streamlit application was developed to provide:

- Dataset Overview
- Data Visualizations
- Model Performance Comparison
- Forecasting Results
- Project Summary

12. Screenshots:

12.1 Home Page:

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Project Overview

This project predicts Tesla stock prices using Deep Learning models.

Features

- Data Analysis
- Data Visualization
- Feature Engineering
- SimpleRNN Model
- LSTM Model
- GridSearchCV Hyperparameter Tuning
- Multi-Step Forecasting
- Model Evaluation
- Performance Comparison

Dataset Rows: 2416 Dataset Columns: 9 Models Used: 2

Description: The Home Dashboard provides an overview of the project, including objectives, key features, and summary statistics of the Tesla stock dataset used for forecasting.

12.2 Dataset Information:

Dataset Information

Dataset Preview

Date	Open	High	Low	Close	Adj. Close	Volume	MA(10)	MA(20)
0 2010-06-29	19	20	17.54	21.89	21.89	21.89	187162300	None
1 2010-06-30	25.79	26.42	23.3	23.83	23.83	173871300	None	None
2 2010-07-01	25	26.50	25.07	21.96	21.96	82188000	None	None
3 2010-07-02	23	23.1	18.71	19.2	19.2	51398000	None	None
4 2010-07-06	20	20	15.83	16.11	16.11	68600000	None	None

Dataset Shape

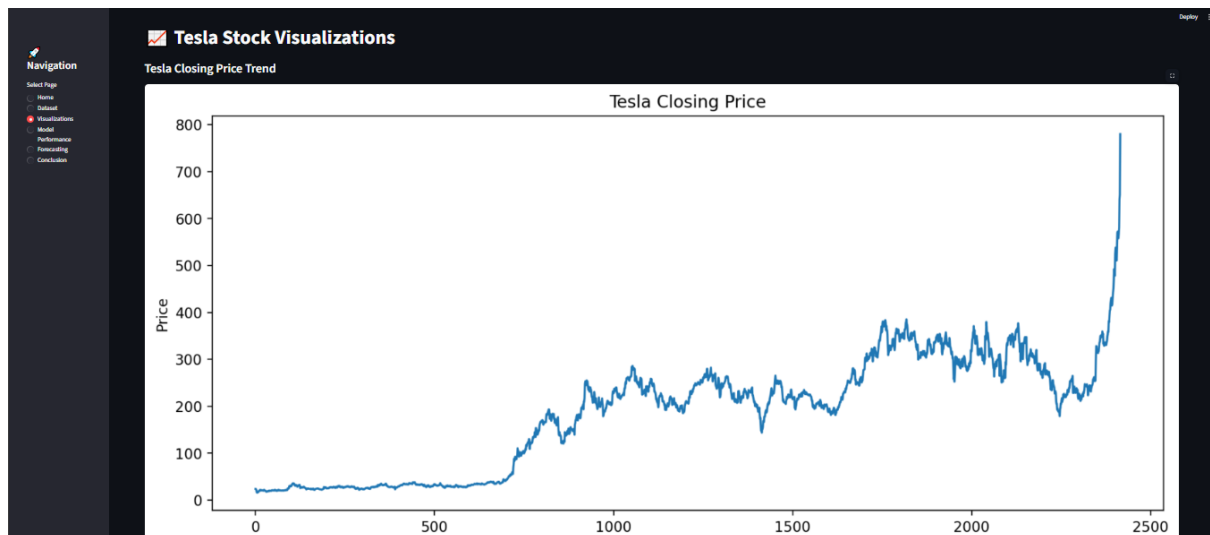
(2416, 9)

Dataset Statistics

	Open	High	Low	Close	Adj. Close	Volume	MA(10)	MA(20)
count	2416	2416	2416	2416	2416	2416	2367	2217
mean	186.2711	189.5782	182.5988	186.4037	186.4037	5573721.6887	184.8745	186.7209
std	118.14027	119.68073	116.45216	119.1136	119.1136	49807609.5339	112.8959	101.6217
min	16.14	16.63	14.58	14.5	14.5	110500	15.672	15.7564
25%	34.34075	34.89075	33.58075	34.4	34.4	18890710	34.7671	46.04
50%	213.885	216.745	208.87	212.96	212.96	4278400	216.8046	222.7024
75%	268.45	270.2175	262.3875	266.775	266.775	73631700	261.2927	267.5817
max	873.69	798.14	673.32	793	793	47865000	443.6032	539.5081

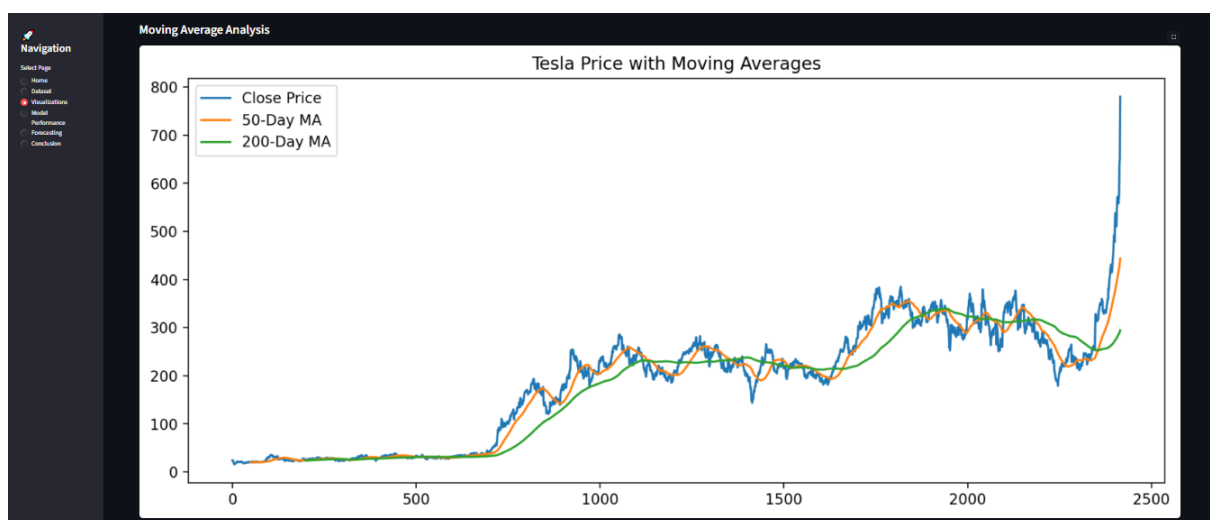
Description: This dashboard displays the dataset preview, shape, and statistical summary, helping to understand the structure and characteristics of Tesla stock market data.

12.3 Tesla Closing Price Trend Visualization:



Description: The visualization shows the historical trend of Tesla's closing stock prices, highlighting significant price movements and long-term growth patterns over time.

12.4 Moving Average Analysis (MA50 & MA200):



Description: The moving average chart compares Tesla's closing prices with 50-day and 200-day moving averages, helping identify market trends and potential trend reversals.

12.5 Model Performance Comparison:



Description: This dashboard compares the performance of SimpleRNN and LSTM models using evaluation metrics such as MSE, RMSE, MAE, and R² Score.

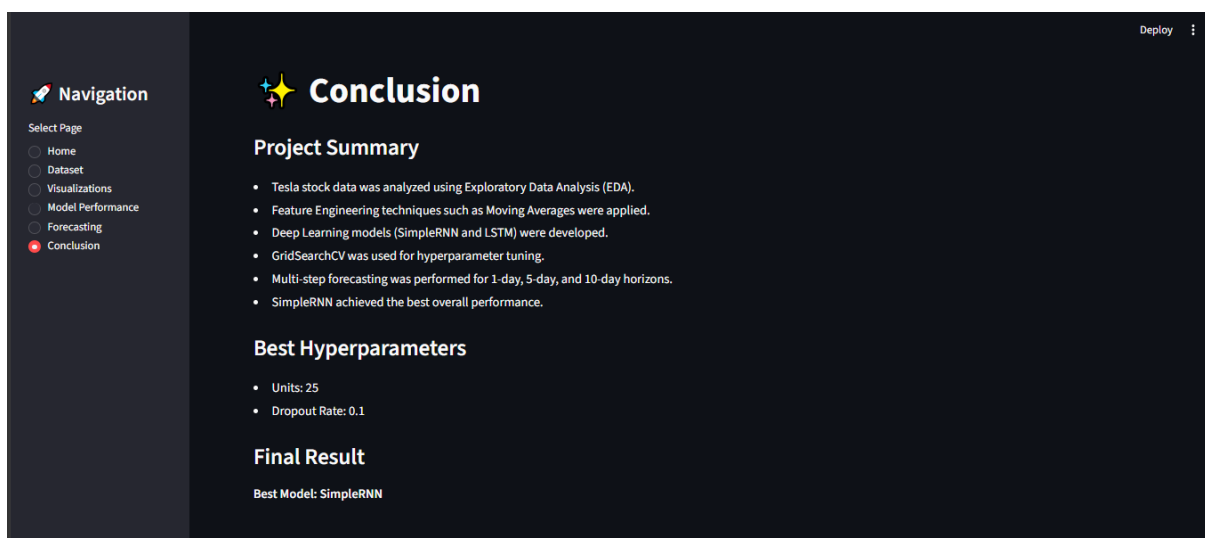
12.6 Multi-Step Forecasting Dashboard:





Description: The forecasting dashboard presents 1-day, 5-day, and 10-day stock price predictions generated by the trained SimpleRNN model, demonstrating its forecasting capability.

12.7 Project Conclusion Dashboard:



Description: The conclusion dashboard summarizes the project outcomes, highlights the best-performing model, and presents the overall findings of the Tesla stock forecasting analysis.

13. Conclusion:

The project successfully demonstrated the application of deep learning techniques for stock market forecasting. SimpleRNN and LSTM models were developed, evaluated, and

compared. Based on evaluation metrics, the SimpleRNN model achieved the best overall performance.

The project further incorporated GridSearchCV-based hyperparameter tuning and multi-step forecasting, making it a comprehensive deep learning solution for Tesla stock price prediction.

14. Future Scope:

- Financial News Sentiment Analysis
 - Macroeconomic Indicator Integration
 - GRU-Based Forecasting
 - Transformer-Based Forecasting
 - ARIMA Comparison
 - Automated Trading Strategy Development
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END OF REPORT